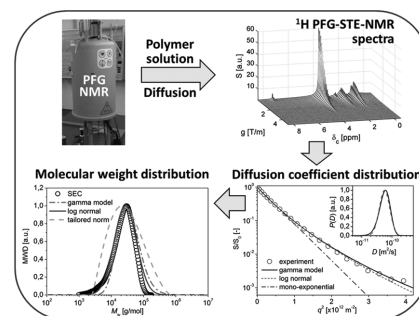


Diffusion in Polymer Solutions: Molecular Weight Distribution by PFG-NMR and Relation to SEC

Xiaoai Guo,* Esther Laryea, Manfred Wilhelm, Burkhard Luy, Hermann Nirschl, Gisela Guthausen

Quantification of diffusion coefficient distribution (DCD) and correlation with molecular weight distribution (MWD) of polymers is still an issue in pulsed field-gradient nuclear magnetic resonance (PFG-NMR). The conventional scaling law utilized so far to relate diffusion coefficient and molecular weight only holds true for the determination of MWD at sufficiently low concentrations. To extend measurement limits and to get a good signal-to-noise ratio, an exponential correlation is introduced to describe the effect of polymer concentration on diffusion in PFG-NMR. Two model polymers (polystyrene and poly(methyl methacrylate)) dissolved in deuterated chloroform are studied at different concentrations in the range of 0.16–8 wt%. The DCDs are determined by modeling the measured signal attenuation with three methods (gamma distribution, log normal distribution, and tailored norm regularization). It is shown that the proposed method applies to the PFG-NMR measurements on polymer solutions over a wide concentration range, providing almost the same MWDs as those obtained at low concentrations. The MWDs retrieved from NMR experiments agree well with those by size exclusion chromatography.



1. Introduction

Knowledge of physical polymer properties, e.g., dynamics in solution, is of great importance to predict the behavior in polymer study and synthesis. In the past decades

various techniques have been developed and utilized for polymer characterization to get a better knowledge of polymer molecular mass, structure, morphology, and diffusion in polymer solutions.^[1,2] Among them, for instance, the molecular weight distribution (MWD) and polydispersity index (PDI) reflect the type of polymerization reaction and thus affect the physical and mechanical properties of polymer mixtures.^[3] Size exclusion chromatography (SEC), also known as gel permeation chromatography, is one of the most common methods used to directly determine MWD based on the hydrodynamic volume of the dissolved polymer. However, SEC measurements require special sample preparation and a large amount of solvent, and they are time consuming. On the other hand, nuclear magnetic resonance (NMR) has proven to be a very useful noninvasive method for characterizing polymer microstructures, understanding polymerization mechanisms,

Dr. X. Guo, Prof. M. Wilhelm
Institute for Chemical Technology and Polymer Chemistry
Karlsruhe Institute of Technology (KIT)
Engesserstrasse 18, 76128 Karlsruhe, Germany
E-mail: xiaoai.guo@kit.edu
Dr. X. Guo, Prof. B. Luy, Prof. H. Nirschl, Prof. G. Guthausen
Pro²NMR IBG-4 and MVM
Karlsruhe Institute of Technology (KIT)
Adenauerring 20b, 76131 Karlsruhe, Germany
E. Laryea
Institute of Thermal Process Engineering
Karlsruhe Institute of Technology (KIT)
Kaiserstrasse 12, 76131 Karlsruhe, Germany

and studying polymer dynamics.^[4,5] Diffusion ordered spectroscopy (DOSY) NMR, i.e., pulsed field-gradient (PFG-) NMR is known to indirectly measure the molecular size via the diffusion coefficient D which is related to molecular weight M . This technique has been used for identification of different chemical components in mixtures via their chemical shifts in many fields from medical and biological sciences to material sciences. A more detailed description of the principles and applications of DOSY can be found in reviews.^[6,7] As compared to SEC, pulsed field-gradient nuclear magnetic resonance (PFG-NMR) has several advantages concerning sample preparation and amount of solvents. Currently, the measurement time is in the same range as SEC. Simultaneous chemical analysis of polymer mixtures is possible via analysis of chemical shift and J -couplings. Moreover, impurities in the polymer solution do not have a direct interference in determining the self-diffusion of the polymeric species if the peak signals of the polymer under study can be resolved in the DOSY spectra, whereas the alternative technique, dynamic light scattering, requires well-prepared dust-free and diluted samples.

PFG-NMR measurements deliver the diffusion coefficient distribution (DCD), which is related to the polymer polydispersity. Historically, only the median values were considered of both, DCD and MWD. The conventional scaling law expressed in the form of $D = KM^{-\alpha}$ has often been used as the basis for the molecular weight measurement by PFG-NMR,^[8] where K and α are scaling parameters depending on the type of the polymer system. Based on this empirical scaling power law, there have been numerous publications related to the influence of molecular weight polydispersity on the PFG-NMR measurements as well as the determination of the average molecular weight and the MWD from the NMR diffusion experiments.^[9–19] It should be pointed out that the scaling parameters (K , α) are only constant for polydisperse systems with the same chemical composition at sufficiently low concentrations. In addition, the actual functional form of the DCD in PFG-NMR experiments is generally unknown, and different distribution functions are explored to describe the signal decay. Early in the 1960s, a single exponential model was derived by Stejskal and Tanner^[20] to model the signal decay for a monodisperse solution with a single self-diffusion coefficient. Later on, Raghavan et al.^[21] studied the self-diffusion of polymers with the Schulz distribution model, which gave reasonably consistent results with those obtained by the single exponential model. For the polymer systems with higher polydispersity, however, a nonexponential signal decay is observed. In this case, the stretched exponential model is often applied to fit the nonexponential signal attenuation curves.^[16,17] As pointed out by Röding et al.,^[22] the relation between the stretch parameter and the spread

of the DCD is complicated, and the stretched exponential model does not correspond to the actual DCD. Apart from the aforementioned models and the numerical inverse Laplace transform (ILT)^[14,23] used in DOSY studies of polydisperse samples, log normal distribution has often been chosen as an alternative to model the experimental signal decay.^[10,15,19] Besides, gamma distribution^[19,22] or gamma convolution models which are convolutions of n ($n \geq 1$) gamma distributions^[24] have recently been derived and applied for data processing in the PFG-NMR studies, providing results similar to those obtained with the log normal model and ILT, respectively. It is worth noting that ILT is often used to extract the DCD from the NMR signal decay but this approach is known for strong vulnerability to noise and numerical instability due to its ill-posed nature. In addition, numerous iterative regularization methods have been intensively studied for determining the DCD, e.g., iterative thresholding algorithm for multi-exponential decay^[25] and tailored norm regularization for monitoring polydispersity by NMR diffusometry.^[26]

To the best of our knowledge, these previous PFG-NMR studies have mainly focused on the discussion on the effects of average molecular weight or polydispersity on the experimental results and the scaling theories, but the MWD was estimated from DCD only at low concentration. In this work, first we investigate and compare the different models for determination of DCD from PFG-NMR data, and second retrieve MWD from DCD under consideration of the concentration dependence of diffusion to extend the measurement limits. The approach results in a correlation between DCD and MWD of polymers in solution and therefore the measurement of MWD by PFG-NMR. Two polymers, namely polystyrene (PS) and poly(methyl methacrylate) (PMMA) are used as test samples dissolved in deuterated chloroform at different concentrations in the range of 0.16–8 wt%. Three data treatment models (gamma distribution, log normal distribution, and tailored norm regularization) are explored. MWDs retrieved from PFG-NMR experiments are compared with those measured by SEC.

2. Experimental Section

2.1. Materials and Sample Preparation

PS and PMMA (Table 1) have been used for PFG-NMR experiments. PS samples (PS1K, PS2K, PS28K and PS125K) were obtained from PSS (Polymer Standards Service, Mainz, Germany), and PS263K was synthesized in house and was kindly provided by Tobias Fischer in Prof. C. Barner-Kowollik's laboratory at KIT. PMMA samples were synthesized by free radical polymerization of MMA, which was carried out in solution at 80 °C in a Taylor–Couette reactor. Xylene

Table 1. Properties of the polymers used for PFG-NMR.

Sample	Weight-averaged molecular weight M_w [kg mol ⁻¹]	Number-averaged molecular weight M_n [kg mol ⁻¹]	Peak molecular weight M_p [kg mol ⁻¹]	Polydispersity index $\bar{D}_M$ [-]
PS1K	0.71	0.64	0.68	1.10
PS2K	1.92	1.77	1.82	1.08
PS28K	27.50	26.60	28.00	1.03
PS125K	125.00	120.00	130.00	1.04
PS263K	263.02	162.48	240.04	1.62
PMMA7K	7.59	4.36	7.99	1.74
PMMA12K	12.09	6.61	11.97	1.83
PMMA29K	28.99	15.80	30.58	1.83
PMMA38K	38.42	21.22	37.46	1.81
PMMA42K	42.45	23.35	49.56	1.82
PMMA52K	52.15	28.82	64.01	1.81
PMMA57K	56.50	34.00	67.41	1.66
PMMA74K	73.90	40.44	80.88	1.83

(mixture of isomers) was used as solvent and 2,2-azoisobutyronitrile as initiator. The initiator concentration was varied from 0.2 to 2 wt% to get samples with different molecular weight distributions. The residual monomer and the solvent of the reaction mixture were eliminated by vacuum evaporation (<25 mbar, 150 °C). Corresponding weight- and number-averaged molecular weights (M_w , M_n) and PDI were determined by the SEC instrument (PL-SEC 50) using tetrahydrofuran as eluent. These polymer samples were dissolved in deuterated chloroform (CDCl₃, 99.5% D, EURISO-TOP, France) at different concentrations in the range of 0.16–8 wt%. The polymer solutions were filled into standard 5 mm NMR tubes (Deutero GmbH, Kastellaun, Germany). To avoid solvent evaporation and thus concentration changes, the filled glass tubes were flame-sealed.

2.2. PFG-NMR Measurements

The ¹H PFG-NMR experiments were performed at 25 °C on a Bruker Avance 200 MHz (Bruker, Germany) NMR spectrometer equipped with a Diff30 diffusion probe, which provides a maximum gradient strength g up to 12 T m⁻¹. The spectra were obtained by using a PFG-STE sequence with 32 linearly spaced gradient steps and 8–16 scans. The PFG-STE parameters for diffusion experiments are: gradient pulse duration $\delta = 2$ ms, diffusion time $\Delta = 40$ ms, and repetition time 5 s. The data were acquired within Topspin 1.5 (Bruker, Germany) and processed via phase correction and integration of the corresponding characteristic peaks in the spectra. Further data processing of the signal decay was done with Matlab routines.

3. Results and Discussion

Figure 1a exemplarily shows the acquired ¹H-PFG-STE spectra for the sample PMMA29K. The peak at $\delta_c = 3.59$ ppm is attributed to the methyl ester group (peak a in Figure 1a). The peaks related to the methyl group and the $-\text{CH}_2-$ group in PMMA are visible at smaller chemical shifts (peak b and peak c in Figure 1a), and the peak of chloroform is also identified at 7.26 ppm in the spectra. The signal intensity S changes with the gradient g . Analysis of the characteristic peak a in the spectra leads to a signal decay curve (Figure 1b). For a monodisperse diluted polymer solution with a single self-diffusion coefficient D , the detected signal decay can be expressed by an exponential function^[20]

$$\frac{S(q)}{S_0} = \exp[-Dq^2(\Delta - \delta/3)] \quad (1)$$

where S_0 is the signal intensity at $q = 0$, and Δ is the diffusion time. The parameter q is defined as the product $\gamma g \delta$, where γ is the proton magnetogyric ratio, g is the gradient amplitude, and δ is the gradient pulse duration.

For a polydisperse solution, the detected signal intensity of a peak in the spectrum is the integral over the NMR-signals of all polymer molecules. Under the assumption that the studied polymers have a certain probability distribution $P(D)$ of molecular weight and conformations, Equation (1) can be rewritten as

$$\frac{S(q)}{S_0} = \int_0^\infty P(D) \exp[-Dq^2(\Delta - \delta/3)] dD \quad (2)$$

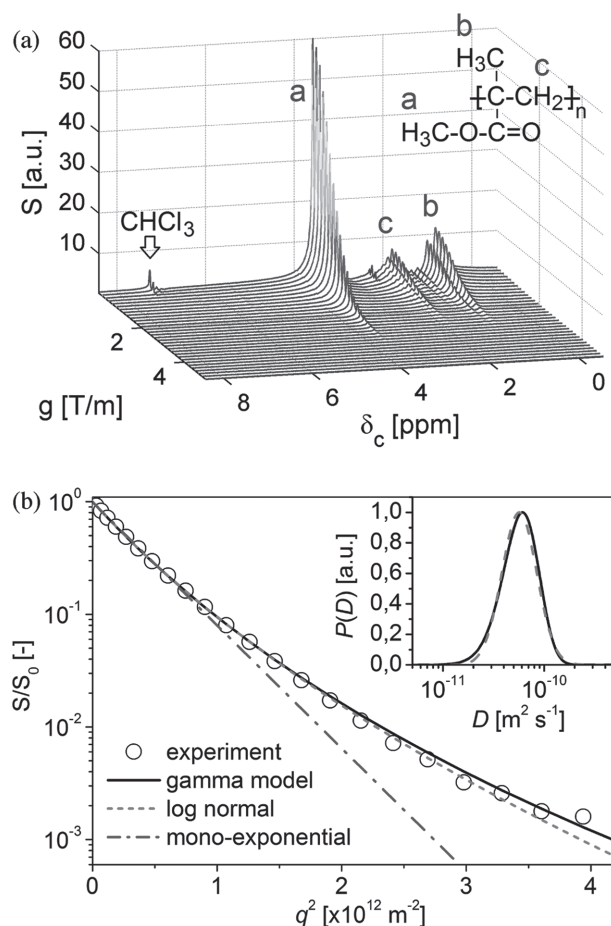


Figure 1. a) ^1H -PFG-STE-NMR spectra for the sample PMMA29K ($c = 3.85$ wt%), b) signal attenuation modeled with a mono-exponential function (dot-dashed line), the gamma distribution model (solid line), and the log normal distribution (dashed line). The inset shows the respective diffusion coefficient distributions extracted with gamma and log normal models, having a mean value of $\approx 7.1 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$. The monoexponential model results in a mean diffusion coefficient of $6.4 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$.

In this work, the DCDs of the polymers are obtained by modeling the acquired PFG-NMR signal decay with log normal distribution $P_{\text{LN}}(D)$ and gamma distribution $P_{\text{G}}(D)$ ^[10,15,19,22]

$$P_{\text{LN}}(D) = \frac{1}{\sqrt{2\pi} D \sigma_{\text{LN}}} \exp \left[-\frac{(\ln D - \ln D_0)^2}{2\sigma_{\text{LN}}^2} \right] \quad (3)$$

with the mean value $D_{\text{mean}} = D_0 \exp(\sigma_{\text{LN}}^2/2)$, where D_0 is the median value and σ_{LN} is a measure of the width of the DCD

$$P_{\text{G}}(D) = \frac{D^{\kappa-1} \exp(-D/\theta)}{\Gamma(\kappa) \theta^{\kappa}} \quad (4)$$

with the mean value $D_{\text{mean}} = \theta \kappa$ and the width of the distribution $\sigma_{\text{G}} = \theta \sqrt{\kappa}$, where θ is a scale parameter, κ is a shape parameter, and Γ is the gamma function.

The distribution parameters of Equations (3) and (4) can be determined by a nonlinear least-squares fitting of Equations (2), (3) and (4) to the observed decays.

The normalized signal attenuation, S/S_0 , was modeled with monoexponential function, gamma model, and log normal distribution (Figure 1b). The monoexponential line describes the data down to approximately $S/S_0 = 10\%$. Below this value, the experimental data deviate from the monoexponential function due to polydispersity. A characteristic of polydisperse solutions is the curvature of S/S_0 observed at larger values for q^2 . Both the gamma and the log normal models describe the experimental data quite well. The difference between the two fit curves is negligibly small on the semilogarithmic scale (Figure 1b) and close to the noise level of the measurement. The corresponding DCDs obtained with these two models are rather similar, too (Figure 1b). A comparison among the experimental results for polymers at different molecular weights indicated that there is a more pronounced attenuation of the signal intensity for the fast diffusing small species, which is reflected by a steeper slope in the semilogarithmic plot. Moreover, with increasing molecular weight, the slope decreases and is indicative for smaller diffusion coefficient. Notably, the signal decay is detected over a wide range of intensity (typically two to three orders of magnitudes) in the ^1H PFG-NMR diffusion experiments.

Figure 2 illustrates the apparent concentration dependence of the DCD for the sample PMMA29K. With increasing polymer concentration, the retrieved DCD curve is shifted to smaller diffusion coefficients (Figure 2a). The characteristic quantities, namely the mean diffusion coefficient D_{mean} , the width of the gamma distribution σ_{G} , and the coefficient of variation (CV = $\sigma_{\text{G}}/D_{\text{mean}}$), are plotted in Figure 2b for the sample PMMA29K at different concentrations ranging from 0.33 to 7.11 wt%, respectively. It is obvious that the mean diffusion coefficient D_{mean} does not change significantly at low concentrations (e.g., $c < 2$ wt%). Increasing the polymer concentration leads to a strong reduction of the mean diffusion coefficient. The width of the distribution σ_{G} shows a similar tendency when the polymer concentration increases. Moreover, the coefficient of variation CV remains nearly constant, which indicates a good model fit at different polymer concentrations. This behavior is, however, of rather unphysical nature but reflects only the limits of the data treatment approaches usually applied. For larger and practically more relevant polymer concentrations, an extension of these approaches is needed.

To further study the influence of macromolecular concentration on the polymer mobility in the given CDCl_3 solutions, diffusion coefficient distributions for different PMMA and PS samples at different concentrations were measured (Figure 3, symbols: D_{mean} and "error bars": σ_{G}).

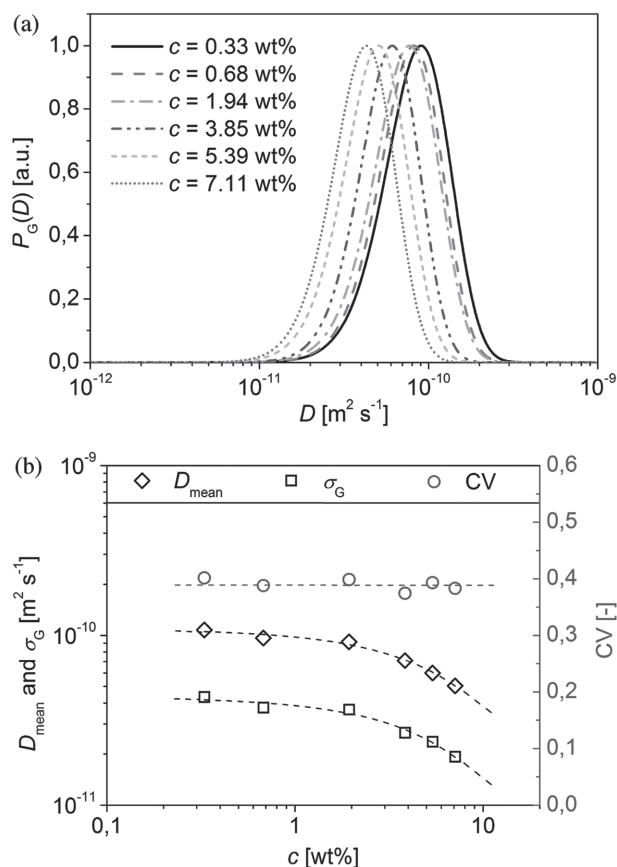


Figure 2. a) NMR determined diffusion coefficient distributions, b) mean diffusion coefficient D_{mean} , width of the distribution σ_G and coefficient of variation ($\text{CV} = \sigma_G / D_{\text{mean}}$) for the sample PMMA29K at different concentrations ($c = 0.33$ wt%, 0.68 wt%, 1.94 wt%, 3.85 wt%, 5.39 wt%, and 7.11 wt%) obtained from a fit to the data with the gamma model.

The PMMA samples with different molecular weights M_w ranging from 7.6 to 74 kg mol^{-1} exhibit very similar curves, showing the same tendency of the concentration dependence in the measured range. Except for a few outliers a similar behavior has been observed for the PS samples with M_w of 0.7 to 263 kg mol^{-1} at different concentrations (Figure 3b). The outliers are probably due to the experimental error and because the PS sample with a very low molecular weight consists of a few monomer units. In the current experimental concentration range, the diffusion coefficients of these oligomers do not show as strong a concentration dependence as those of the others. Besides, the chosen concentrations left a gap at low concentrations for PS2K not present in the others. As compared to the previous study on diffusion of PS in CDCl_3 ,^[27] in which the mono-/biexponential model only gave apparent diffusion coefficients, now both gamma and log normal models describe the data better and provide more information about the polymer diffusion in solution. Furthermore, these parameters, i.e., the mean

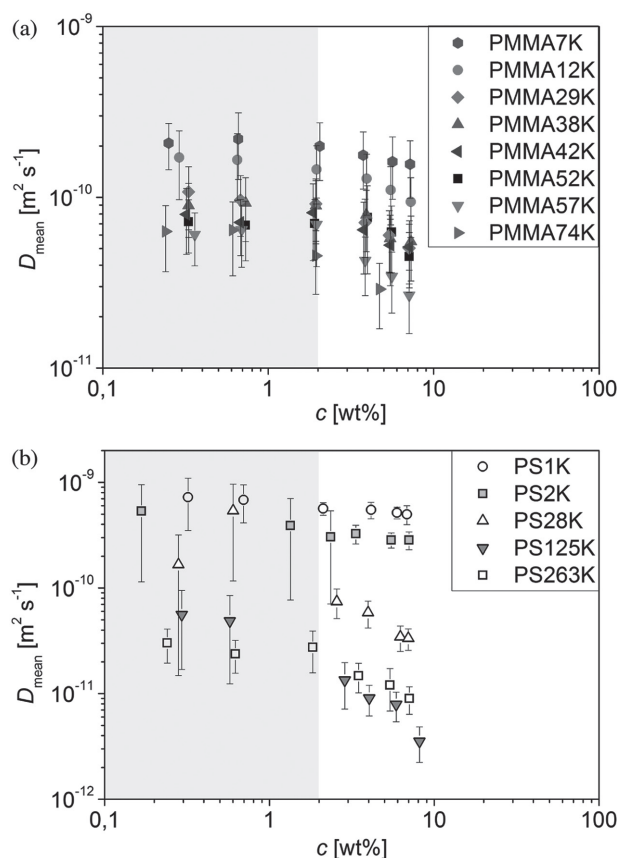


Figure 3. Diffusion coefficient distributions at different concentrations (Symbols: D_{mean} , "error bars": width of the distributions), a) PMMA samples with molecular weights M_w ranging from 7.6 to 74 kg mol^{-1} , b) PS samples with M_w of 0.7 to 263 kg mol^{-1} .

diffusion coefficient D_{mean} , the width of the distribution σ_G , and the coefficient of variation (CV) can be related to the polydispersity. As derived by Rödinger et al.,^[22] the PDI $\mathcal{D}_M$ for the gamma model is given by

$$\mathcal{D}_M = \left(1 + \frac{\sigma_G^2}{D_{\text{mean}}^2} \right)^{\frac{1}{\alpha^2}} = (1 + \text{CV}^2)^{\frac{1}{\alpha^2}} \quad (5)$$

where α is an empirical scaling parameter, depending on the type and quality of the solvent and the monomer. With the given PDI values from SEC and the distribution parameters for the studied polymer samples in CDCl_3 , the parameter α in Equation (5) was estimated to be 0.51 ± 0.05 . The scaling parameter and the extension of the conventional scaling law to a wide concentration range for extracting MWD from DCD will be discussed below.

A further comparison between the DCDs for PMMA and PS samples obtained using the gamma model and the log normal model is illustrated in Figure 4, where the solid line with a slope of one reflects the perfect correlation between two methods. It is obvious that almost all the data points of the mean diffusion coefficient D

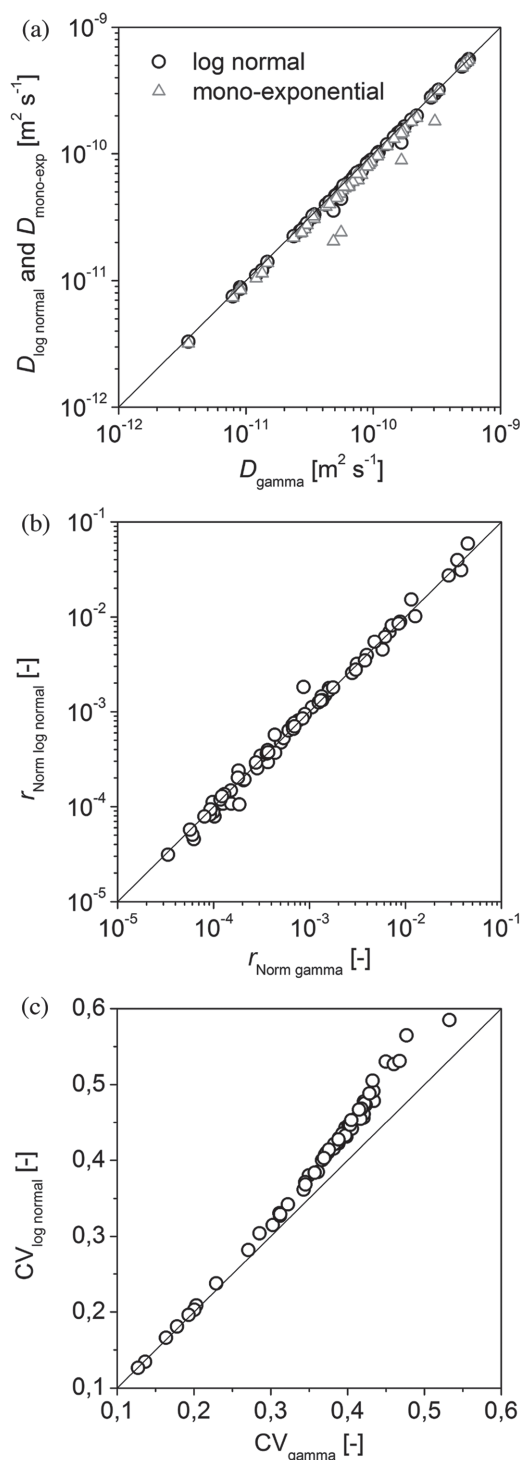


Figure 4. Comparison of DCDs obtained from the gamma model and the log normal model, a) diffusion coefficients D_{mean} , in comparison with D achieved with the monoexponential function, b) squared 2-norm of the residuals r_{Norm} , c) coefficient of variation CV.

(circular dots in Figure 4a) lie on the solid line, implying that the results obtained with both models correlate well. The diffusion coefficients D_{monoexp} achieved with a fit of

a monoexponential function are also plotted for comparison. They are systematically underestimated, resulting in an approximate estimation in contrast to those obtained with gamma and log normal models. This can be attributed to the distribution of diffusion coefficients being obvious in deviation of the measured signal decay from the straight line (Figure 1b). The squared 2-norm of the residuals r_{Norm} and the coefficients of variation CV are also close to the solid line (Figure 4b,c), reflecting the similarity and goodness of fit. The values of CV for the gamma model are somewhat smaller than those for log normal model (Figure 4c). This suggests that the gamma model has somewhat smaller residuals relative to the predicted values, and thus provides a better fit to the experimental data for PMMA and PS samples in CDCl_3 solutions in the measured polymer concentration range up to ≈ 8 wt%. A similar behavior has also been observed in the previous study on other polymer solutions, e.g., polyethylene glycol in D_2O ^[22] and PS in CDCl_3 solutions^[19] at very low polymer concentrations ($c \leq 0.1$ wt%). Notably, although in the present work the PFG-NMR experiments were carried out at the concentration up to 8 wt%, the concentration range can be further extended upward for determining DCD.

The above discussion shows that DCDs of the polymer solutions can be determined by modeling and analyzing the PFG-NMR signals. The experimental signal attenuation depends on the diffusion of the polymer molecular species present in the solution, which is directly related to the molecular size and polydispersity. Thus, the PFG-NMR measurement provides the possibility to retrieve the MWD from the experimental DCD. For this purpose, the well-known empirical scaling law has often been used to establish the relation between molecular weight M_w and mean self-diffusion coefficient D for dilute solutions, typically expressed as^[8]

$$D = K \cdot M_w^{-\alpha} \quad (6)$$

where K and α are scaling parameters depending on the type of the studied polymer system. This power law only holds true for sufficiently low polymer concentrations.

Figure 5 shows the experimental scaling between diffusion coefficient D and molecular weight M_w for PS and PMMA samples in CDCl_3 dilute solutions ($c \approx 0.3$ wt%) measured with PFG-NMR. The solid line represents a nonlinear least-squares fit of the experimental data. The scaling $D = 2.19 \times 10^{-8} M_w^{-0.52}$ was found. The value of the scaling parameter ($\alpha = 0.52$) agrees fairly well with the values in the range of $\alpha = 0.47$ – 0.61 for PS and PMMA in CDCl_3 ^[28,29] as well as in previous NMR diffusion studies of other polymer solutions such as $\alpha = 0.52$ – 0.55 for poly(ethylene oxide) (PEO) in D_2O ,^[15] $\alpha = 0.55$ for PEO in water,^[30] $\alpha = 0.46$ – 0.54 for PMMA in acetone.^[28] It must be noted that the mapping of MWD onto DCD lacks a firm theoretical background and thus needs to be treated

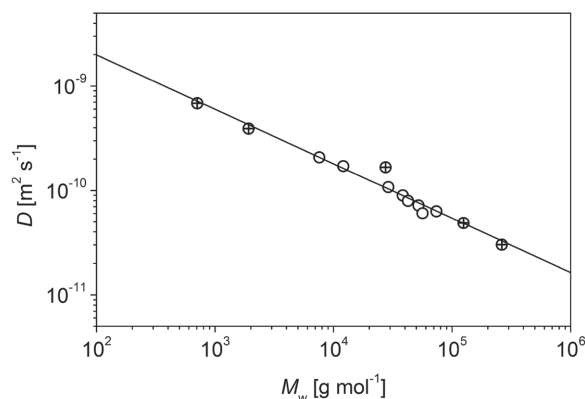


Figure 5. Scaling between mean diffusion coefficient D and molecular weight M_w for PS samples ($\oplus$) and PMMA samples ($\circ$) in CDCl_3 dilute solutions ($c \approx 0.3$ wt%) measured with PFG-NMR.

carefully. For dilute polymer systems, this scaling relation can give accurate prediction of the MWD of polymers with known molecular weights.^[15,28]

Using this scaling relationship between diffusion coefficient D and molecular weight M_w , the diffusion coefficient distribution $P(D)$ can be transformed into the weight-averaged molecular weight distribution $P_M(M)$ or the number-averaged molecular weight distribution $P_N(M)$

$$P_M(M) = P_N(M)M \quad (7)$$

$$P_M(M) = P(D) dD / dM \quad (8)$$

With Equations (6) and (8), the molecular weight distribution function $P_M(M)M$ can be determined as

$$P_M(M)M = \alpha D P(D) = \alpha K M^{-\alpha} P(D) \quad (9)$$

$\mathcal{D}_M$ is defined as the ratio of the medians of the weight-averaged molecular weight $\bar{M}_w$ to the number-averaged molecular weight $\bar{M}_n$

$$\mathcal{D}_M = \frac{\bar{M}_w}{\bar{M}_n} = \frac{\int P_M(M) M dM / \int P_M(M) dM}{\int P_N(M) M dM / \int P_N(M) dM} \quad (10)$$

The diffusion coefficient distribution $P(D)$ used for retrieving the molecular weight distribution can be either a specific distribution function in closed form (e.g., Schulz distribution, gamma distribution, log normal distribution) or a general distribution function without analytical form (e.g., a DCD function obtained from the signal decay with numerical algorithms as ILT).

The molecular weight distributions for PMMA29K and PS263K in CDCl_3 dilute solutions ($c \approx 0.3$ wt%) retrieved from PFG-NMR DCDs show very small differences between different methods in Figure 6 (PMMA29K: $\mathcal{D}_M$ 1.91 for the gamma model, and 1.87 for the log normal model; PS263K: $\mathcal{D}_M$ 1.65 for the gamma model, and 1.66

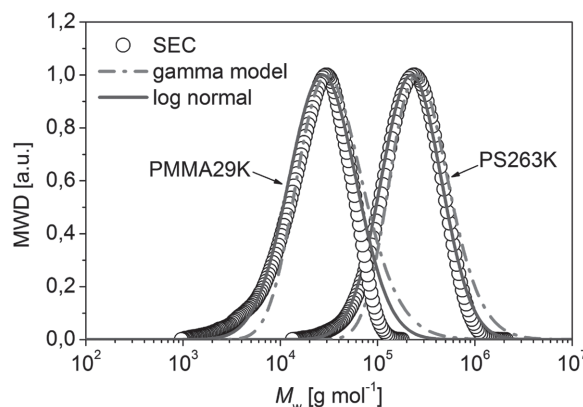


Figure 6. Molecular weight distributions for PMMA29K and PS263K in CDCl_3 dilute solutions ($c \approx 0.3$ wt%) retrieved from the DCDs determined with the gamma model and the log normal model, and compared with SEC results.

for the log normal model). Furthermore, these MWDs agree well with those by SEC (PMMA29K: $\mathcal{D}_M$ 1.83; PS263K: $\mathcal{D}_M$ 1.62) (Figure 7). The parameters of MWDs for

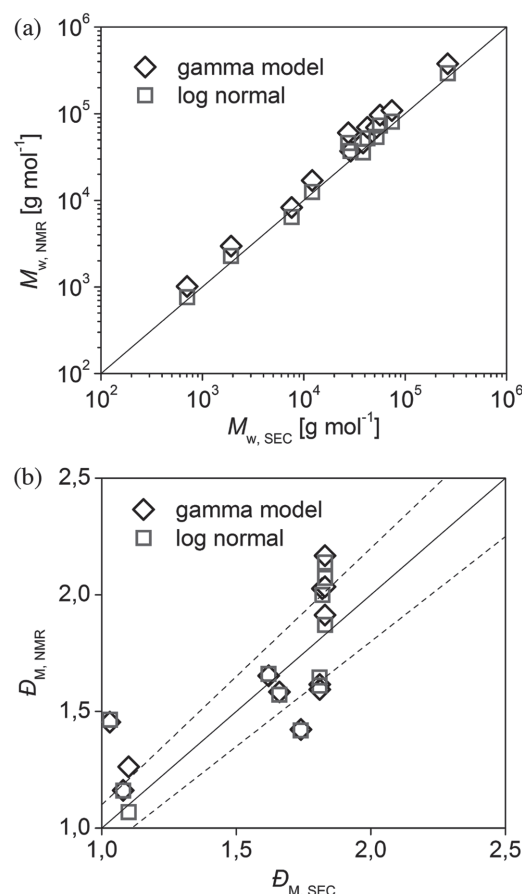


Figure 7. Comparison between M_w and PDI retrieved from NMR diffusion measurements on PMMA and PS samples in CDCl_3 dilute solutions ($c \approx 0.3$ wt%) and those obtained by SEC, a) weight-average molecular weight M_w , b) PDI $\mathcal{D}_M$, two dashed lines indicate a range with a PDI variation of $\pm 10\%$.

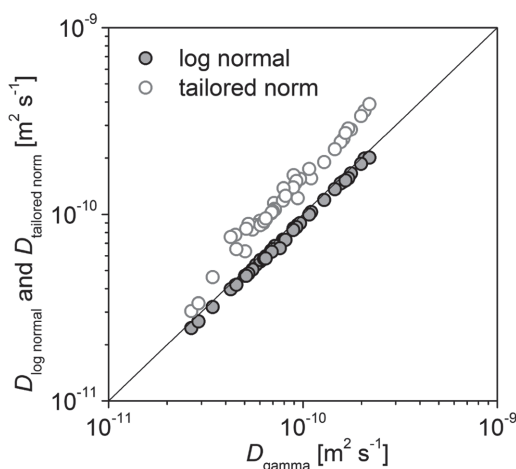


Figure 8. Comparison between diffusion coefficients by tailored norm regularization and those by gamma and log normal models.

the other samples measured by PFG-NMR and SEC are also illustrated in Figure 7. It is obvious that in the measured range of $M_w = 0.71\text{--}263\text{ kg mol}^{-1}$ the molecular weights M_w determined by PFG-NMR are well correlated with the SEC results, although the results using the gamma model show a slight overestimation as compared to those using the log normal model in Figure 7a. In addition, the PDI values determined from PFG-NMR also show a good agreement with those by SEC. Most of the measurement data lie in the range with a PDI variation of $\approx \pm 10\%$ except for only a few points probably due to measurement errors or deviations of the real distribution shape from the used models. The overall agreement between the results by PFG-NMR and SEC is quite satisfactory.

Besides the gamma and log normal models for processing the signal decay data, a new iterative regularization method called tailored norm regularization^[26] has also been explored for determining the DCD and thus the MWD. It is based on the use of ℓ_p -norm ($1 \leq p \leq 2$) as a regularization for ILT. Compared to the above-discussed gamma and log normal models, however, the tailored norm method results in an overestimation of the mean diffusion coefficients for the studied samples (Figure 8). This could be due to the iterative method itself and the imperfect presetting of some parameters in the available iteration program apart from the inherent sensitivity to noise and numerical instabilities which are well known for ILT. The retrieved DCD is broader, leading to a broader MWD than those by gamma and log normal models as well as by SEC. As an example, MWDs retrieved with tailored norm regularization method, gamma, and log normal models for the sample PMMA29K in CDCl_3 dilute solution ($c \approx 0.3\text{ wt\%}$) as well as by SEC are compared in Figure 9. In conclusion, both the gamma and log normal models utilized for the data treatment in PFG-NMR on polymers in solution are capable of describing the

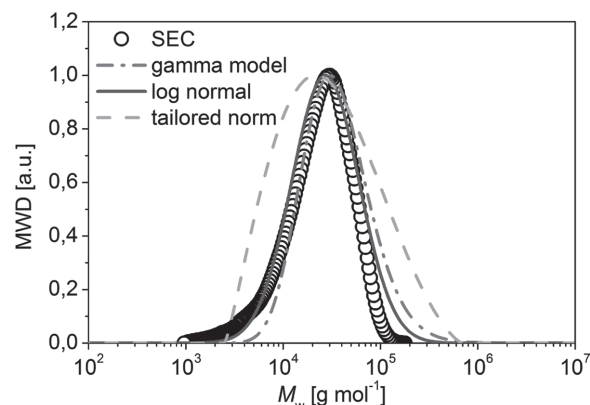


Figure 9. MWDs retrieved with tailored norm regularization method, gamma, and log normal models for PMMA29K in CDCl_3 ($c \approx 0.3\text{ wt\%}$) as well as by SEC.

experimental data quite well and result in a good prediction of DCDs and thus MWDs of polymer solutions.

As already stated (Figures 2 and 3), the apparent diffusion coefficient depends on the concentration of the polymer solution. The mean diffusion coefficient varies slightly or remains almost constant at low concentrations. The diffusion coefficient decreases with increasing the concentration because of the interaction of polymer molecules in the solution. Masaro and Zhu^[31] reviewed different physical models of diffusion in polymer solutions, gels, and solids as well as their limitations. The underlying physical concepts are obstruction effects, hydrodynamic interactions, and free volume theory, but their applicability varies widely. Considering these different concepts and the experimental observations in this work, it is found that an exponential equation can be employed for good description of the diffusion variation with the concentration in present PFG-NMR experiments. Thus, the concentration dependent diffusion coefficient in terms of the stretched exponential equation^[32,33] can be simplified to Equation (11)

$$D = D_0 \exp(-kc^v) \xrightarrow{\text{for } v=1} D = D_0 \exp(-kc) \quad (11)$$

where D_0 is the diffusion coefficient at very low concentrations, c is the concentration of the polymer solution, v is the stretch parameter depending on the type of polymer solutions, and k is a proportionality constant obtained by a nonlinear least-squares fit on the diffusion data. This model has a simple form and shows a good correlation with the experimental data (Figure 10a). Gao and Fagerness^[34] have employed this model to study the diffusion in hydroxypropylmethyl cellulose gels and determine the drug and water diffusivity by NMR.

To our knowledge, MWD is estimated from DCD based on the scaling law only at very low concentration so far. To circumvent this limitation and to further extend the

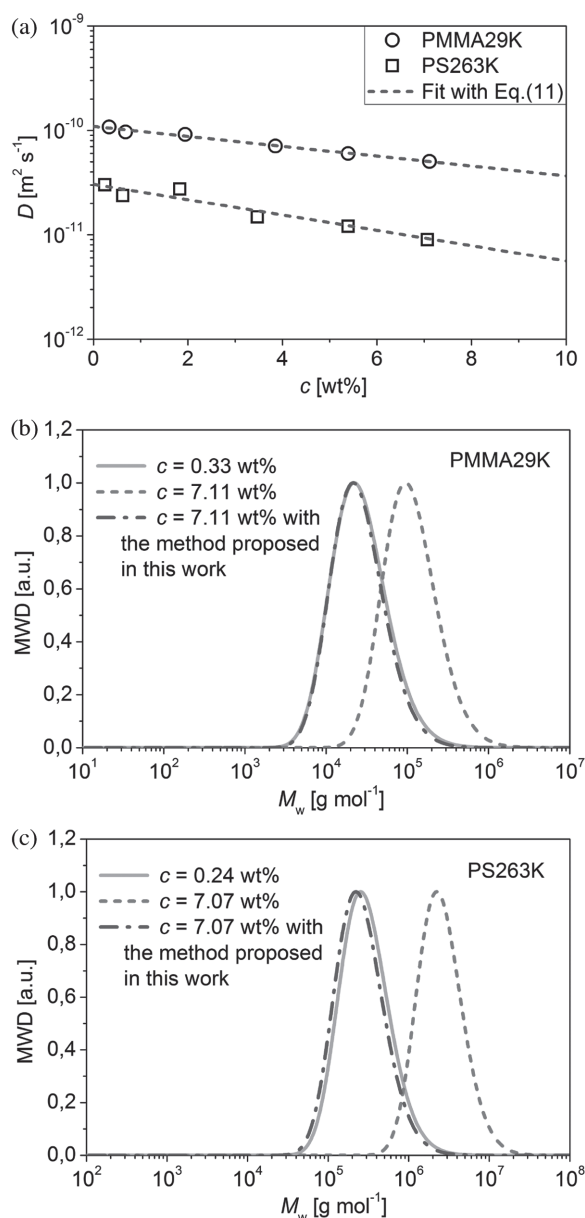


Figure 10. a) $D_{\text{mean, gamma}}$ of PMMA29K and PS263K as a function of concentration (cf. Figures 2 and 3), the fit with Equation (11) leads to $k = 10.9$ [–] and 16.9 [–] for PMMA29K and PS263K, respectively, b) molecular weight distributions retrieved from DCDs determined from the PFG-NMR measurements on PMMA29K at different concentrations with and without applying the correction by Equation (11), c) corresponding results for PS263K at different concentrations.

measuring range, a new facile method is introduced for retrieving MWD from DCD for a broader concentration range. As discussed, at very low concentration the scaling law in Equation (6) applies the form of $D_0 = K \cdot M^{-\alpha}$. Thus, Equation (11) can be rewritten as

$$D = K \exp(-kc) M^{-\alpha} \quad (12)$$

MWD $P_M(M)M$ can be correlated with the diffusion coefficient distribution $P(D)$ determined from PFG-NMR at different concentrations

$$P_M(M)M = \alpha K \exp(-kc) M^{-\alpha} P(D) \quad (13)$$

Figure 10 exemplarily depicts the diffusion coefficients of PMMA29K and PS263K as a function of concentration and the MWDs retrieved with the method proposed above. The experimental diffusion coefficients at the measured concentrations can be well described with Equation (11) (Figure 10a). The trend of diffusion with increasing polymer concentration is similar. Besides, DCDs determined from PFG-NMR diffusion measurements at high concentration have been used to retrieve MWDs. Without concentration correction, the resulting MWDs at high concentration deviate significantly from that at low concentration (cf. dashed line and solid line in Figure 10b,c). Considering the concentration effect, the MWDs agree perfectly well with that at low concentration (cf. dot-dashed line and solid line in Figure 10b,c). Similar results have been obtained for the other molecular weight samples at high concentrations. In summary, PFG-NMR is very suitable for simultaneous quantification of DCDs and MWDs of polymers. The measurements can also be extended to high concentration, greatly improving the signal-to-noise ratio which improves accuracy and applicability of PFG-NMR. Finally, it should be pointed out that, apart from the studied polymer solutions (PS and PMMA dissolved in CDCl_3), the proposed method in terms of Equations (11) and (13) for determination of MWDs at different concentrations can be generalized for further application to other polymer solutions as follows

$$P_M(M)M = \alpha K \exp(-kc^v) M^{-\alpha} P(D) \quad (14)$$

The stretch parameter v can be determined experimentally, depending on the type of polymer solutions under study.

4. Conclusions

Self-diffusion of two polymers in solution (PMMA and PS in CDCl_3) has been investigated at different concentrations in the range of 0.16–8 wt% by means of the PFG-NMR. DCDs have been determined by modeling the signal decay with different approaches, namely the gamma model, the log normal model and the numerical tailored norm regularization. Experimental results indicate that both the gamma and log normal models describe the diffusion data very well, giving DCDs comparable to SEC, whereas the tailored norm method results in broader DCDs. Based on the scaling between diffusion coefficient and molecular weight, MWDs of the studied polymers have been retrieved

from DCDs. Furthermore, the known dependence of the diffusion coefficient on concentration above a certain value leads to an overestimation of the MWD when the conventional scaling law is employed. To release this limitation, an exponential function of polymer concentration is introduced into the conventional scaling law for consideration of the influence of concentration on self-diffusion. The MWDs derived with the proposed method agree perfectly well with those measured at low concentrations, resulting in a very satisfactory calculation of MWDs of polymers. The retrieved MWDs agree well with those by SEC. As a result, PFG-NMR is a very good analytical tool for simultaneous quantification of DCDs and MWDs of polymers and chemical identification of polymers by spectral analysis. The measurements can now also be extended to higher and realistic concentrations inherently improving the signal-to-noise ratio, i.e., the accuracy and therefore the applicability of the PFG-NMR in the study of polymer solutions.

Acknowledgements: The authors gratefully acknowledge the financial support from the German Research Foundation (DFG SFB 1176 Project Q2 as well as Pro²NMR instrumental facility at KIT and RWTH Aachen). B.L. acknowledges support by the HGF program BIFTM. The authors also thank Tobias Fischer in Prof. C. Barner-Kowollik's laboratory at KIT for kindly providing the sample PS263K.

Received: September 12, 2016; Revised: October 18, 2016;
Published online: November 18, 2016; DOI: 10.1002/macp.201600440

Keywords: diffusion; diffusion coefficient distribution; molecular weight distribution; polymer solutions; pulsed field-gradient NMR

- [1] R. Kimmich, *NMR—Tomography Diffusometry Relaxometry*, Springer Verlag, Berlin 1997.
- [2] D. Campbell, R. A. Pethrick, J. R. White, *Polymer Characterization: Physical Techniques*, 2nd ed., Stanley Thorne, Cheltenham, UK 2000.
- [3] M. Rogosic, H. J. Mencer, Z. Gomzi, *Eur. Polym. J.* **1996**, 32, 1337.
- [4] F. A. Bovey, P. A. Mirau, *NMR of Polymers*, Academic Press, San Diego 1996.
- [5] R. Kimmich, N. Fatkullin, in *Advances in Polymer Science* (Eds: A. Abe, A.-C. Albertsson, K. Dusek, W. H. de Jeu, S. Kobayashi, K.-S. Lee, L. Leibler, T. E. Long, I. Manners, M. Möller, E. M. Terentjev, B. Voit, G. Wegner, U. Wiesner, M. J. Vicent, J. Genzer), Springer Verlag, Berlin Heidelberg **2004**, p. 1.
- [6] C. S. Johnson, *Prog. Nucl. Magn. Reson. Spectrosc.* **1999**, 34, 203.
- [7] Y. Cohen, L. Avram, L. Frish, *Angew. Chem. Int. Ed.* **2005**, 44, 520.
- [8] P. G. Gennes, *Scaling Concepts in Polymer Physics*, Cornell University Press, Ithaca 1979.
- [9] P. T. Callaghan, D. N. Pinder, *Macromolecules* **1981**, 14, 1334.
- [10] P. T. Callaghan, D. N. Pinder, *Macromolecules* **1985**, 18, 373.
- [11] G. Fleischer, *Polymer* **1985**, 26, 1677.
- [12] H. Walderhaug, F. K. Hansen, S. Abrahmsen, K. Persson, P. Stilbs, *J. Phys. Chem.* **1993**, 97, 8336.
- [13] H. Walderhaug, O. Soderman, D. Topgaard, *Prog. Nucl. Magn. Reson. Spectrosc.* **2010**, 56, 406.
- [14] A. Chen, D. H. Wu, C. S. Johnson, *J. Am. Chem. Soc.* **1995**, 117, 7965.
- [15] B. Hakansson, M. Nyden, O. Söderman, *Colloid Polym. Sci.* **2000**, 278, 399.
- [16] X. L. Gong, E. W. Hansen, Q. Chen, *Macromol. Chem. Phys.* **2011**, 212, 1007.
- [17] X. L. Gong, E. W. Hansen, Q. Chen, *Macromol. Chem. Phys.* **2012**, 213, 278.
- [18] N. E. Kuz'mina, S. V. Moiseev, V. I. Krylov, V. A. Yashkir, V. A. Merkulov, *J. Anal. Chem.* **2014**, 69, 953.
- [19] N. H. Williamson, M. Nyden, M. Röding, *J. Magn. Reson.* **2016**, 267, 54.
- [20] E. O. Stejskal, J. E. Tanner, *J. Chem. Phys.* **1965**, 42, 288.
- [21] R. Raghavan, T. L. Maver, F. D. Blum, *Macromolecules* **1987**, 20, 814.
- [22] M. Röding, D. Bernin, J. Jonasson, A. Sarkka, D. Topgaard, M. Rudemo, M. Nyden, *J. Magn. Reson.* **2012**, 222, 105.
- [23] S. W. Provencher, *Comput. Phys. Commun.* **1982**, 27, 229.
- [24] M. Röding, N. H. Williamson, M. Nyden, *J. Magn. Reson.* **2015**, 261, 6.
- [25] M. Urbanczyk, D. Bernin, W. Kozminski, K. Kazimierczuk, *Anal. Chem.* **2013**, 85, 1828.
- [26] M. Urbanczyk, D. Bernin, A. Czuron, K. Kazimierczuk, *Analyst* **2016**, 141, 1745.
- [27] M. Cudaj, J. Cudaj, T. Hofe, L. Luy, M. Wilhelm, G. Guthausen, *Macromol. Chem. Phys.* **2012**, 213, 1833.
- [28] S. Auge, P. O. Schmit, C. A. Crutchfield, M. T. Islam, D. J. Harris, E. Durand, M. Clemancey, A. A. Quoineaud, J. M. Lancelin, Y. Prigent, F. Taulelle, M. A. Delsuc, *J. Phys. Chem. B* **2009**, 113, 1914.
- [29] G. Fleischer, O. E. Zgadzai, V. D. Skirda, A. I. Maklakov, *Colloid Polym. Sci.* **1988**, 266, 201.
- [30] K. Chari, B. Antalek, J. Minter, *Phys. Rev. Lett.* **1995**, 74, 3624.
- [31] L. Masaro, X. X. Zhu, *Prog. Polym. Sci.* **1999**, 24, 731.
- [32] G. D. J. Phillies, *Macromolecules* **1986**, 19, 2367.
- [33] G. D. J. Phillies, *Macromolecules* **1987**, 20, 558.
- [34] P. Gao, P. E. Fagerness, *Pharm. Res.* **1995**, 12, 955.